

Shallow Neural Networks for Signal Decomposition

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Abstract—This thesis compares shallow Conv1D-based neural networks with deeper baselines for supervised signal decomposition. Thirteen architectures were evaluated and compared on synthetic mixtures in the main benchmark. TasNet performed best, reaching a macro correlation of 0.887 and macro SNR of 8.97 dB. The best shallow model reached a lower macro correlation of 0.811, but used only 78,070 parameters compared with 2.5 million for TasNet. The results show that shallow models cannot fully replace deeper architectures for maximum accuracy, but they offer a practical alternative when model size, latency, or computational cost are more important. In short, the best shallow model trades a 0.076 drop in macro correlation for a 32 times reduction in parameters and a 2.6 times lower inference latency, making it attractive whenever model size, latency, or compute budget dominate the deployment decision.

Git Repository: <https://github.com/VBortsov/SD-SNN-Thesis>

Index Terms—signal decomposition, shallow neural networks, Conv1D, temporal convolutional network, source separation, model efficiency

I. INTRODUCTION

Signal decomposition, the process of separating a mixture into meaningful components such as oscillations, trends, or independent sources, is central to biomedical engineering [1], [14], structural health monitoring [25], audio processing [2], [15], and communications [12]. Deep learning models achieve strong performance on this task [2], [9], [15], [22], but their high parameter counts make them impractical for embedded or real-time applications. This thesis investigates whether shallow neural networks can provide a competitive, resource-efficient alternative.

A. Classical Methods

Classical methods are efficient but rely on assumptions or manual parameters. The Fourier [12] and wavelet [5] transforms suit stationary and non-stationary signals respectively, while adaptive methods such as EMD [6] and VMD [13] reduce the need for fixed bases at the cost of noise sensitivity or manual mode-count selection.

B. Deep Neural Networks

Neural networks learn complex mappings directly from data. Time-domain models such as TasNet [2], Demucs [19], and SepFormer [15], and wider surveys of waveform-domain

methods [20], encoder-decoders such as U-Net [9], and iterative residual networks such as IRCNN⁺ [22] have all demonstrated strong decomposition performance. Their main drawback is a high parameter count, often in the millions, which limits deployment on embedded or real-time systems.

C. Shallow Neural Networks

Shallow neural networks (SNNs) trade modelling capacity for speed and deployability [7], [8]. Simple 1-D CNNs have proven effective for ECG analysis, vibration monitoring, and audio classification [1], and their low inference time suits real-time embedded use. Their main limitation is a restricted receptive field: a single convolutional layer captures only local patterns, making it harder to separate components that interact across time scales. Therefore, during this study we targeted architectural additions that can reduce this gap without sacrificing efficiency.

D. Problem Statement and Research Questions

It is unclear whether deep-model accuracy can be approximated by much smaller, shallower networks for signal decomposition. Three research questions guide this thesis:

- **RQ1.** Can SNNs achieve acceptable decomposition performance while using significantly fewer parameters and less computation?
- **RQ2.** What are the fundamental limitations of SNNs, and under what conditions does the performance gap become most pronounced?
- **RQ3.** Which architectural modifications contribute most to improving shallow models, and how far can they close the gap to deep networks?

To answer these, 13 models are compared on a supervised synthetic decomposition task using quality metrics (macro correlation, macro SNR, test loss) and efficiency metrics (parameter count, inference time, training time). The main results of this thesis are:

- A controlled comparison of thirteen architectures for supervised signal decomposition using one shared training and evaluation pipeline. These architectures cover a parameter range of around two orders of magnitude.
- An incremental ablation study that shows which shallow model modifications improve performance. The results indicate that the receptive-field width is more important than the total number of parameters.

This thesis was prepared in partial fulfilment of the requirements for the Degree of Bachelor of Science in Data Science and Artificial Intelligence, Maastricht University. Supervisors: [Pietro Bonizzi, Iris Huijben]

- An accuracy-efficiency analysis, including a deployability check on real EEG data. This analysis shows in which situations a compact 1-D CNN can be preferred over a deeper separator.

II. METHODS

A. Models Compared

Thirteen models were evaluated in total and grouped into deep neural network baselines and shallow alternatives. In every model, all components are predicted at once in a single forward pass. For this study, a model is classified as an SNN if it has fewer than 100,000 parameters and no deep multi-stage separator (such as a TCN separator stack or dual-path transformer); all other models are classified as DNNs. The 100,000-parameter threshold is not fundamental; it is chosen because it cleanly separates the single- or few-layer Conv1D family (all below 80,000 parameters here) from the multi-stage separators (all above 600,000), leaving a wide margin around the boundary so that the classification is insensitive to its exact value. Full layer-by-layer architectures for all SNN are given in Appendix B. *Deep baselines.* TasNet [2] is a time-domain encoder-separator-decoder model in which a temporal convolutional network estimates component masks in a learned latent space. It was chosen as the primary deep baseline because it is a widely used, high-performing time-domain separation architecture that operates directly on raw waveforms, which matches the setting of this study; its separator stack uses eight residual dilated-convolution blocks (see Appendix B). UNet1D adapts the U-Net architecture [9] to one-dimensional signals, using successive downsampling and skip connections to preserve fine-grained temporal features, and is included as a representative encoder-decoder baseline. SepFormer [15] processes signals with a dual-path transformer [26], capturing both local and global dependencies through multi-head self-attention [4], and represents the attention-based family. The TasNet, U-Net, and SepFormer baselines were reimplemented in PyTorch following the published descriptions, adapting input/output dimensions to the decomposition task; no third-party reference implementation was used, so all models share the same training and evaluation pipeline. *Shallow baselines.* A SingleHead MLP served as the simplest possible baseline: a fully connected network with no explicit temporal structure, mapping the flattened input directly to the output components. The base shallow architecture was then a Classic Conv1D network, which applies a single convolutional layer with a fixed kernel size to capture local temporal patterns, followed by a linear output head. This single-layer Conv1D was chosen as the base because it is the minimal architecture that still introduces an explicit temporal (translation-invariant) inductive bias, making it the natural starting point for measuring the effect of each added modification. A convolutional autoencoder was also included as a shallow baseline: it compresses the input into a bottleneck and reconstructs components through a symmetric decoder, with fewer than 25,000 parameters and no learned latent masking. All remaining shallow models are

architectural variants of the Classic Conv1D base, described in the following subsection.

B. Improving the Shallow Model

All modifications use the Classic Conv1D network as their baseline and extend it via optional components that can be switched on or off independently. The evaluated shallow models are specific on/off combinations of these components; the exact module sets are shown in Fig. 3a and Appendix B. The incremental evolution from the Classic Conv1D baseline to the final model is illustrated in Appendix C. Nine modifications were investigated, covering both architectural changes and a training-objective change:

- 1) **Multi-scale convolutional branches.** Parallel Conv1D branches with different kernel sizes (e.g. 5, 7, 9) capture local patterns at multiple temporal resolutions simultaneously [3].
- 2) **Dilated convolutions.** Gaps are introduced between filter taps to widen the receptive field without adding layers or parameters [3]. Different dilation rates across branches cover a range of temporal scales.
- 3) **TCN blocks.** Standard Conv1D layers are replaced by Temporal Convolutional Network blocks [10], which combine dilated causal convolutions with residual connections for structured receptive-field growth.
- 4) **Feature fusion.** Branch outputs are merged via either a 1×1 convolution (learned linear combination) or bilinear fusion (pairwise cross-branch interactions).
- 5) **Multiple output heads.** Each decomposed component receives a dedicated prediction head, encouraging component-specific specialisation and reducing inter-component interference.
- 6) **Attention stem.** A small convolutional stem extracts initial features, and a lightweight channel-attention mechanism [28] weights the branch outputs to emphasise the most informative temporal scale per input.
- 7) **Frequency-aware input features.** The raw signal is concatenated with its DFT magnitude spectrum [12]:

$$\tilde{x} = \text{concat}(x, |X[k]|), \quad X[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi kn/N}. \quad (1)$$

This provides explicit spectral information for components that overlap in time but differ in frequency.

- 8) **Residual connections.** Skip connections between convolutional layers improve gradient flow and allow low-level temporal features to be reused at later stages without increasing depth [10].
- 9) **Component-specific loss.** Adds per-component and spectral penalties to the global reconstruction loss:

$$\begin{aligned} \mathcal{L} = & \lambda_c \frac{1}{K} \sum_{k=1}^K \text{MSE}(\hat{y}_k, y_k) + \lambda_r \text{MSE}\left(\sum_{k=1}^K \hat{y}_k, x\right) \\ & + \lambda_s \frac{1}{K} \sum_{k=1}^K \|\mathcal{F}(\hat{y}_k) - \mathcal{F}(y_k)\|_1, \end{aligned} \quad (2)$$

where $\text{MSE}(a, b) = \frac{1}{T} \sum_t (a_t - b_t)^2$, $\mathcal{F}(\cdot)$ is the DFT magnitude, y_k and $\hat{y}_k$ are the ground-truth and predicted k -th components, and $(\lambda_c, \lambda_r, \lambda_s) = (1.0, 0.5, 0.05)$ were set empirically on the validation set.

C. Datasets

All experiments use synthetically generated signals, where the true underlying components are known by construction. Each signal is 1024 samples at 256 Hz (4 s). Components are normalised and combined as a weighted sum with randomly sampled per-component weights; additive Gaussian noise ($\sigma \sim \mathcal{U}[0, 0.05]$) is applied to the mixture. Two dataset configurations were used. The 3-component baseline consists of:

- **Harmonic:** a superposition of 1–3 sinusoids with amplitudes from [0.2, 1.0], frequencies from [2, 40] Hz, and random phases.
- **AM-FM:** a signal with instantaneous frequency varying sinusoidally around 8 Hz (± 6 Hz) and a slow amplitude envelope (0.1–1.0 Hz modulation).
- **Chirp:** a polynomial chirp of random order $P \in \{1, 2, 3\}$, starting frequency $f_0 \in [3, 15]$ Hz, with instantaneous frequency $f(t) = f_0 + \sum_{p=1}^P (p+1)a_p t^p$ varying polynomially over the window.

The 5-component extended study adds:

- **Trend:** a zero-mean polynomial (up to cubic) with amplitude in [0.25, 0.8], representing slow non-oscillatory drift.
- **Transient:** a Gaussian pulse with randomised centre (15–85% of duration), width (2.5–9%), and amplitude in [0.4, 1.0].

The full training pool contains 8000 samples. All models were also trained on five sub-100% fractions (5%, 10%, 25%, 50%, 75%), giving six training-set sizes in total (including 100%); each is an independent run from scratch. A fixed validation and test set were held out across all experiments. Since component order in model output is not guaranteed, evaluation uses permutation-invariant matching, re-assigning predicted components to ground truth by minimising total MSE.

D. Training Setup

All models were trained with the Adam optimiser [11] ($\text{lr} = 10^{-3}$, weight decay = 10^{-5}) for 20 epochs, saving the checkpoint with the lowest validation loss. The loss is the combined objective defined in (2). Batch size was 32 for all models except TasNet (16) and SepFormer (8) due to memory constraints. All experiments were run on an NVIDIA RTX 4080M and an Intel Core i9-13900HX.

E. Evaluation Metrics

A full suite of 16 signal-level metrics was computed per model, spanning amplitude-domain error (MSE, RMSE, MAE), scale-invariant separation quality (SI-SDR [23], macro SNR), shape similarity (macro/per-component Pearson correlation), and frequency-domain fidelity (spectral convergence,

log spectral distance). The full list is in Appendix D. Three complementary primary metrics are reported in the main tables; the rest are consistent with these and appear only in the appendix. **Test loss** (MSE) is the direct training objective and measures raw reconstruction error. **Macro correlation** is the mean Pearson r across components, scale-invariant and directly reflecting component shape recovery, which is the core decomposition goal. **Macro SNR** captures residual energy relative to the true component:

$$\text{SNR} = 10 \log_{10} \left(\frac{\sum_t y_t^2}{\sum_t (y_t - \hat{y}_t)^2} \right). \quad (3)$$

This provides an interpretable absolute quality scale consistent with standard source-separation practice [24]. For efficiency comparison, parameter count and inference time were recorded per model. All metrics were computed after permutation-invariant matching [27], as described in Section II-C.

III. EXPERIMENTS

Five experiments address the three research questions, all using the setup in Section II-D and a fixed held-out test set. **E1: Main benchmark.** All 13 models trained on the full 3-component dataset; quality and efficiency metrics compared (RQ1). **E2: Ablation study.** Shallow modifications added incrementally from the Classic Conv1D base to identify which changes contribute most (RQ3). **E3: Data efficiency.** All models retrained at six data fractions (5%–100%) to assess convergence speed and whether SNNs are more data-efficient (RQ1, RQ2). **E4: Real-world EEG.** Best deep and shallow models applied to CHB-MIT scalp EEG [14] (recordings from pediatric patients with intractable seizures), using reconstruction proxy metrics since no ground-truth decomposition is available. **E5: Failure mode analysis.** Component predictions classified into failure categories (shape failure, amplitude mismatch, suppression, overprediction, poor separation) with signal-difficulty descriptors to identify conditions driving the SNN–DNN gap (RQ2).

IV. RESULTS

All supervised metrics are computed on the fixed held-out synthetic test set after permutation-invariant matching. The EEG experiment uses proxy metrics (Section IV-D) since no ground-truth decomposition exists for real recordings.

A. Main Benchmark Comparison and Computational Cost

Table I shows TasNet as the top performer (macro correlation 0.887, SNR 8.97 dB). The best shallow model, Attention-Stem Multi-Head Multiscale TCN, reached 0.811 correlation and 4.52 dB SNR with only 78,070 parameters (a $32.1\times$ reduction from TasNet’s 2.5M) and 23.51 ms inference versus 60.90 ms. All pairwise differences vs. TasNet are statistically significant ($p < 0.05$, paired permutation test, 20,000 iterations). Fig. 1 visualises the Pareto frontier: the Attention-Stem Multi-Head Multiscale TCN surpasses SepFormer in accuracy while using far fewer parameters, making it the strongest candidate for deployment-constrained settings. Fig. 2 confirms

TABLE I
MAIN BENCHMARK CI 95% RESULTS ON THE FULL 3-COMPONENT SYNTHETIC DATASET.

Model	Type	Params	Loss	Corr.	95% CI	SNR (dB)	Inf. (ms)	Train
TasNet	DNN	2,503,296	0.132	0.887	[0.860, 0.913]	8.97	60.90	167.9 min
Attention-Stem Multi-Head Multiscale TCN	SNN	78,070	0.171	0.811	[0.766, 0.857]	4.52	23.51	81.2 min
UNet1D	DNN	2,710,819	0.159	0.778	[0.733, 0.820]	4.21	21.70	67.4 min
Attention-Stem Multi-Head Multiscale Branches	SNN	17,638	0.225	0.777	[0.739, 0.816]	3.97	5.77	30.7 min
SepFormer	DNN	601,088	0.184	0.727	[0.692, 0.761]	5.721	13.14	66.0 min
Attention-Stem Bilinear-Fusion Multi-Head Multiscale Branches	SNN	29,302	0.217	0.693	[0.628, 0.760]	3.14	9.38	41.4 min
Shallow Conv1D	SNN	5,443	0.294	0.668	[0.634, 0.703]	2.46	1.85	10.4 min
Autoencoder	SNN	23,651	0.145	0.660	[0.617, 0.701]	3.08	57.72	40.1 min
Multiple-Head Multiscale Branches	SNN	5,859	0.241	0.659	[0.614, 0.701]	2.14	4.47	19.7 min
Multiscale Dilated	SNN	12,867	0.276	0.644	[0.609, 0.680]	2.10	3.06	10.5 min
Multiscale Branches	SNN	5,859	0.241	0.643	[0.602, 0.684]	1.77	3.42	17.3 min
SingleHead MLP	SNN	643	0.395	0.557	[0.544, 0.569]	1.35	0.32	11.6 min
Fuse Shallow	SNN	5,859	0.238	0.545	[0.491, 0.599]	0.96	3.16	9.4 min

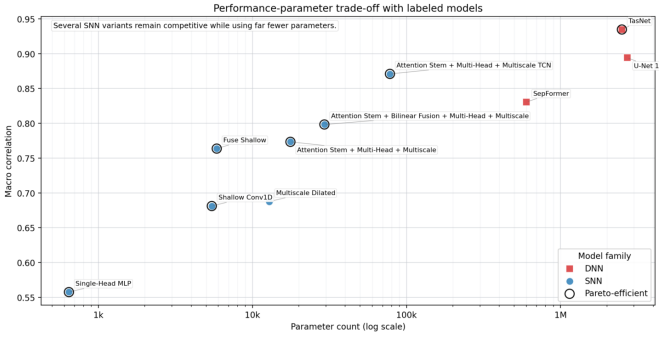


Fig. 1. Performance-parameter trade-off between macro correlation and parameter count (x-axis logarithmic). Beyond the numbers in Table I, this plot shows which models are dominated and which sit on the frontier, and exposes the roughly two-orders-of-magnitude parameter gap between the best shallow model and the deep baselines.

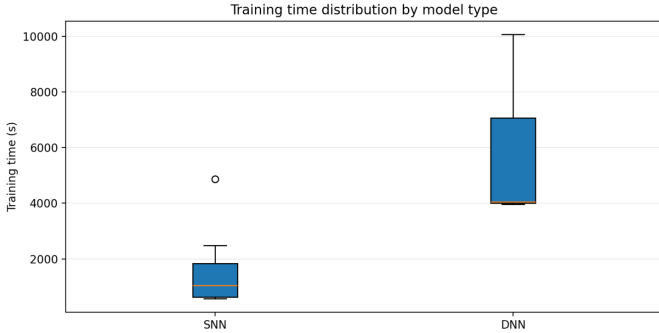


Fig. 2. Training time distribution by model type.

that DNNs are also more expensive to train: TasNet is the largest outlier, and the SNN group has a lower median and smaller spread.

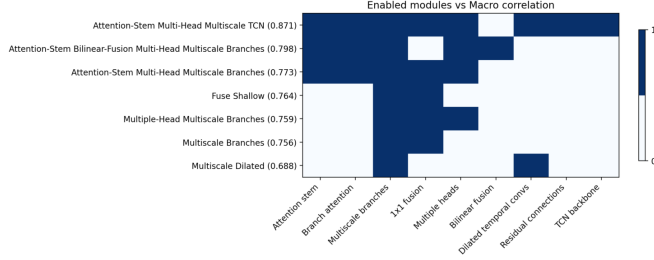
B. Shallow Model Ablation Study

Fig. 3a shows the exact module set enabled for each evaluated shallow variant (see also Appendix B). Fig. 3a shows that the strongest shallow variant combines an attention stem, multi-scale branches, multiple output heads,

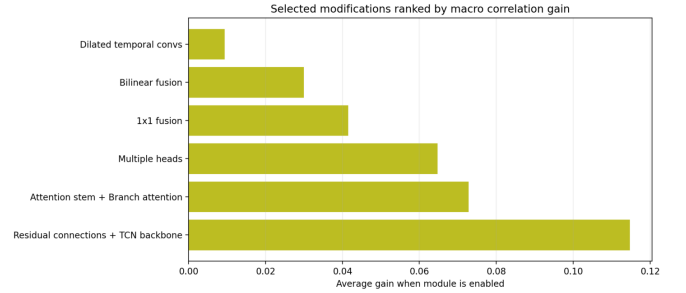
residual connections, and a TCN backbone, reaching a macro correlation of 0.871 in the seed-testing view. The next-best variants achieve 0.798 and 0.773, respectively. This pattern suggests that the performance improvement is not caused by a single isolated component, but by combining temporal multi-scale processing with component-specific prediction and a stronger temporal backbone. The 0.871 figure (ablation seed-testing view) vs. 0.811 (main benchmark) reflects run-to-run variance: the ablation reports the best seed, whereas the benchmark uses a paired permutation test. The benchmark value is the more conservative and comparable figure used throughout. Fig. 3b ranks selected modifications by average macro-correlation gain. The largest gain is associated with residual connections combined with a TCN backbone, with an average gain of 0.114. Attention stem with branch attention also provides a substantial gain of approximately 0.073, followed by multiple output heads at approximately 0.065. Fusion mechanisms provide smaller but still positive gains, while dilated temporal convolutions alone contribute only a minor average gain. These results indicate that the most effective shallow improvements are those that increase temporal modelling capacity while preserving a compact architecture. Multi-scale processing and multiple output heads help the model specialise across components, but the largest improvement comes from adding a stronger temporal backbone. Therefore, shallow models can be improved meaningfully, but the best-performing shallow architecture is already moving toward a more structured temporal model.

C. Data Efficiency Study

Models were trained on increasing fractions of the synthetic training set and evaluated on the same fixed test set. Fig. 4 shows the DNN average rising faster and reaching a higher peak: the SNN–DNN gap widens with data, indicating that deep models benefit more from additional training samples while shallow models saturate earlier. Fig. 5 shows individual curves: TasNet rises from ≈ 0.73 to 0.887, the shallow TCN saturates below the best deep models, and simpler shallow variants show little gain. At intermediate fractions UNet1D occasionally ranks above the shallow TCN due to run-to-run



(a) Enabled shallow-model modules and resulting macro correlation.



(b) Selected modifications ranked by average macro-correlation gain, from smallest gain at the top to the largest at the bottom.

Fig. 3. Shallow-model ablation results. Blue cells indicate enabled modules, while the bar chart ranks selected modifications by their average macro-correlation gain when enabled, ordered from the most to the least contributing modification (top to bottom).

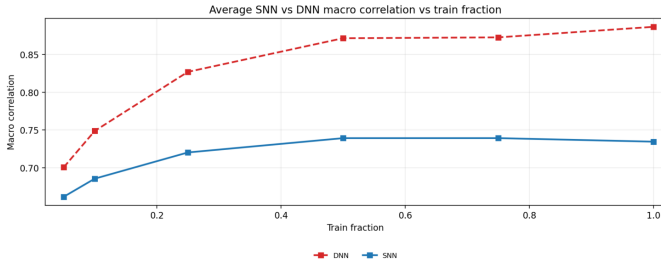


Fig. 4. Average macro correlation for SNNs and DNNs across training-data fractions.

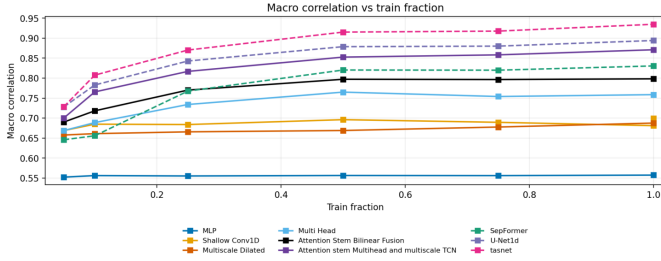


Fig. 5. Macro correlation versus training-data fraction for individual models.

variance, but the ordering at 100% matches Table I. Overall, SNNs reach reasonable performance with limited data but DNNs make better use of larger datasets.

D. Real-World EEG Generalisation and Deployment Cost

The best deep and shallow models are applied to CHB-MIT EEG windows [14]. Since no ground-truth decomposition exists for real recordings, quality is assessed via reconstruction proxies (residual MSE, reconstruction correlation, explained variance of the *summed* output against the observed signal) alongside deployment metrics (latency, model size, RTF). These proxies measure how well predicted components sum back to the input, not how well the underlying physiology is recovered, and are interpreted as consistency and deployability checks. Fig. 6 shows the real EEG reconstruction behaviour of TasNet and the best shallow model. TasNet tracks the observed EEG signal more closely and produces a smaller residual.

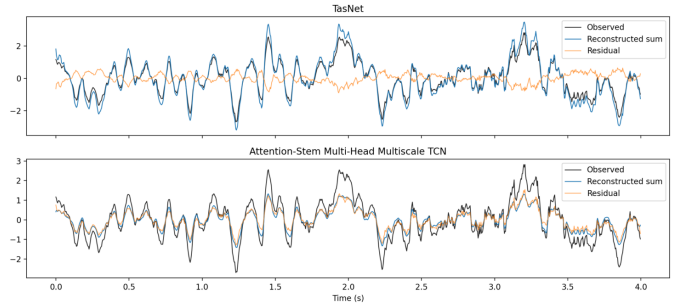


Fig. 6. Real EEG reconstruction comparison for TasNet and the Attention-Stem Multi-Head Multiscale TCN. Each panel shows the observed signal, reconstructed sum, and residual.

The shallow TCN-based model follows the broad waveform structure, but the reconstructed sum is smoother and underestimates several high-amplitude peaks. This indicates that the shallow model transfers to real EEG signals, but with a larger reconstruction error than TasNet. Table II confirms the quality-cost trade-off: TasNet leads on reconstruction fidelity (residual MSE 0.074, explained variance 0.926) while the shallow TCN scores 0.232 and 0.768 respectively, but is $3.4\times$ faster (26.15 vs. 88.50 ms), $32.1\times$ smaller in parameters, and $30\times$ smaller on disk (0.97 vs. 28.79 MB). Both models run well below real-time (RTF <0.03). This real-data experiment highlights the main practical limitation of DNN-based decomposition. TasNet gives better reconstruction behaviour on real EEG, but it requires a larger model and slower inference. For offline analysis, this cost may be acceptable. For real-time monitoring, wearable EEG systems, or edge deployment, the lower latency and smaller memory footprint of the shallow model may be more important than the additional reconstruction quality provided by the deep model. The 0.97 MB checkpoint of the shallow model is small enough for microcontroller-class devices, while TasNet at 28.79 MB would require external memory on most embedded targets. In practice, a tiered setup seems reasonable: the shallow model runs on-device for continuous screening, and flagged windows are sent to TasNet for more detailed offline analysis.

TABLE II

REAL EEG PROXY METRICS AND DEPLOYMENT-COST COMPARISON ON CHB-MIT WINDOWS. **CKPT. MB** IS THE SIZE OF THE SAVED MODEL CHECKPOINT ON DISK IN MEGABYTES. **RTF** IS THE REAL-TIME FACTOR, THE PROCESSING TIME DIVIDED BY THE DURATION OF THE SIGNAL WINDOW (HERE 4 S); **RTF** < 1 MEANS THE MODEL RUNS FASTER THAN REAL TIME. **EXPL. VAR.** IS THE EXPLAINED VARIANCE OF THE SUMMED RECONSTRUCTION, AND **SPECTRAL OVERLAP** IS THE MEAN PAIRWISE SPECTRAL OVERLAP BETWEEN PREDICTED COMPONENTS.

Model	Params	Ckpt. MB	Latency (ms)	RTF	Residual MSE	Recon. corr.	Expl. var.	Spectral overlap
TasNet	2,503,296	28.79	88.50	0.0221	0.074	0.995	0.926	0.785
Attention-Stem Multi-Head Multiscale TCN	78,070	0.97	26.15	0.0065	0.232	0.994	0.768	0.618

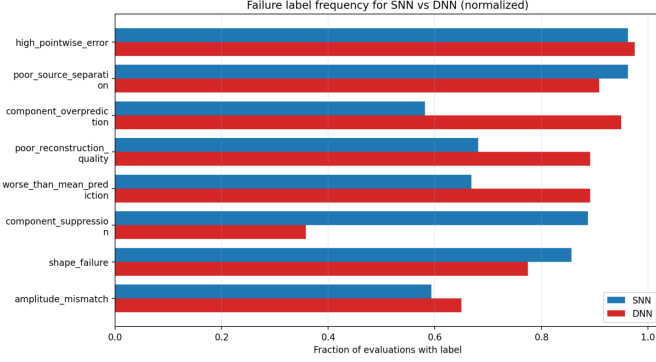


Fig. 7. Normalized averaged failure-label frequency for the shallow models and DNNs. Values indicate the fraction of evaluated components assigned to each label.

TABLE III

COMPONENT-LEVEL RESULTS FOR THE HARD CASE (TWO AM-FM COMPONENTS; **AMFM_2** IS THE LOW-AMPLITUDE ONE). METRICS AFTER PERMUTATION-INVARIANT MATCHING. THE **BEST MODEL** COLUMN INDICATES WHICH MODEL ACHIEVES HIGHER PEARSON CORRELATION ON THAT COMPONENT.

Component	Corr. SNN	Corr. DNN	SNR SNN	SNR DNN	Best model
amfm_1	0.920	0.970	3.532	6.089	TasNet
chirp	0.992	0.995	6.365	4.009	TasNet
amfm_2	0.148	0.107	-12.501	-2.626	SNN TCN

E. Failure Mode Analysis

Each component prediction was matched to its ground-truth counterpart and classified into one or more non-exclusive failure labels: shape failure, amplitude mismatch, suppression, overprediction, poor source separation, poor reconstruction, high pointwise error, and worse-than-mean prediction. The frequency plot in Fig. 7 shows that both model families fail on difficult examples, but with different patterns. The shallow model shows higher component suppression and shape failure, suggesting that it more often misses weak or difficult components. DNNs show higher component overprediction and poor reconstruction labels in this diagnostic set, indicating that their errors are more often related to adding excess energy or structure. Table III summarises the component-level metrics for the representative hard case shown afterwards. Correlation and SNR are reported after component matching. The hard-case example in Fig. 8 shows that the shallow TCN can follow the broad oscillatory structure for some components,

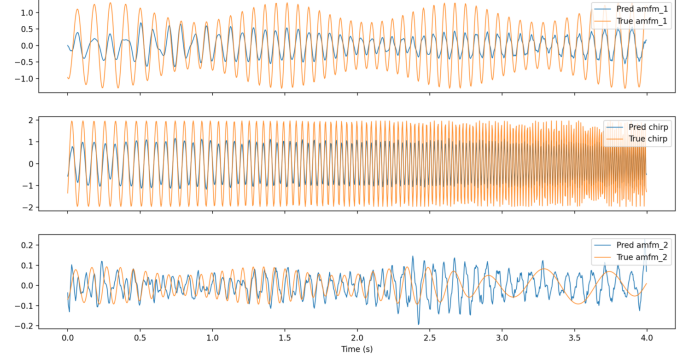


Fig. 8. Hard case reconstruction. Mixture signal, summed reconstruction, and residual are shown; per-component decompositions for both models are given in Appendix E.

but it underestimates amplitude and struggles with fine temporal detail. The AM-FM component with low amplitude is especially difficult, showing poor shape recovery and low SNR. This supports the main conclusion from the quantitative results: shallow models can approximate the decomposition in easier cases, but their limitations become more visible when components are weak, strongly modulated, or spectrally similar. In summary, the failure mode analysis explains where the shallow-deep performance gap comes from. The gap is not uniform across all signals; it is concentrated in difficult examples involving weak components, amplitude imbalance, source overlap, or non-stationary temporal structure.

V. DISCUSSION

A. Accuracy-Efficiency Trade-Off

DNNs provide the best decomposition quality [8], consistent with strong results from TasNet [2], U-Net [9], and attention-based separators [4], but at higher model and training cost. The question is not whether SNNs can replace DNNs, but when their efficiency advantage outweighs the accuracy gap: offline analysis can absorb TasNet’s cost, while real-time, wearable, or repeated-training settings favour compact 1D CNNs [1].

B. Interpretation of Shallow-Model Modifications

The ablation study shows that shallow performance depends more on temporal inductive bias than on parameter count alone. The SingleHead MLP performed poorly because it has no explicit local temporal structure, while Conv1D-based variants improved substantially. The most useful additions

were residual connections with a TCN backbone, attention-based scale selection, multi-scale branches, and multiple output heads. This is consistent with prior work on dilated convolutions for larger receptive fields [3] and TCNs for sequence modelling [10]. Looking across the ablation results, the modifications with the largest gains (TCN backbone, dilation, multi-scale branches) all increase the effective temporal receptive field, while those that only add parameters without extending temporal context (e.g. fusion variants) gave smaller improvements. This points to receptive field width as the main bottleneck for shallow models at this signal length. When adapting a shallow Conv1D to a new task, it appears more effective to invest capacity in mechanisms that widen temporal context per parameter than to simply increase layer width or depth. The largest single ablation gain came from the residual dilated TCN backbone, which is the same temporal primitive at the heart of Conv-TasNet [21]. Conv-TasNet replaced the recurrent separator of TasNet [2] with dilated 1-D convolutions and residual connections; the strongest shallow model here is a compact analogue of that separator core at roughly 78,000 parameters rather than several million, differing in that it regresses components directly and adds multi-scale branches and an attention stem. That an efficiency-driven search over shallow designs converged on this same primitive suggests that receptive field width, not raw depth, is the primary driver of time-domain decomposition quality.

C. Data Efficiency and Real-Data Behaviour

DNNs benefit more from additional data [8], so SNNs are most attractive under data, compute, or time constraints. The CHB-MIT experiment [14] is a proxy check of reconstruction consistency and deployment cost rather than a supervised accuracy result; TasNet reconstructed real EEG more closely, while the shallow TCN was smaller and faster, consistent with the synthetic benchmark trade-off.

D. Failure Modes

The two families fail differently: the shallow model shows more suppression and shape failure (missing weak components due to its limited receptive field), while DNNs show more overprediction and pointwise error (adding excess energy). The hard-case AM-FM example supports this: the shallow TCN recovers the dominant component reasonably well but collapses on the low-amplitude one. Deep models are more robust under difficult conditions (weak, overlapping, or non-stationary components) at the cost of size and inference time; shallow models are preferable when approximate decomposition is acceptable.

E. Societal and Clinical Implications

The efficiency profile of the shallow model has practical consequences for biomedical deployment. A 0.97 MB model that runs well below real time on modest hardware can perform continuous on-device screening of physiological signals such as EEG, which keeps raw recordings local to the sensor and reduces the privacy exposure inherent in streaming patient data

to remote servers. The lower compute cost also lowers the energy footprint per inference, which matters both for battery-limited wearables and for the cumulative cost of always-on monitoring at scale. I stress that these are deployability and consistency arguments, not a clinical validation: any screening use would require physiologically grounded evaluation and regulatory review before it could inform diagnosis. The tiered setup proposed in Section IV-D: a shallow model on-device for continuous screening, with flagged windows escalated to a deep model offline - is one concrete way to combine the privacy and energy benefits of the compact model with the higher fidelity of the deep one.

F. Limitations and Future Work

Supervised evaluation relies on synthetic data; real-world signals have richer noise and non-stationarity. The CHB-MIT results are a consistency and deployability check on out-of-distribution signals, not proof of physiologically correct decomposition. Future work should compare against compressed DNNs (pruned [16], quantised [18], distilled [17]). The more precise deployment question is which point on the accuracy–cost frontier each approach reaches for a fixed parameter or latency budget. Semi-synthetic evaluation (artificial components added to real EEG backgrounds) would provide measurable ground truth under realistic conditions. Inference benchmarking on CPU or embedded hardware and multi-seed reporting would also strengthen the conclusions.

VI. CONCLUSION

Shallow Conv1D-based models achieve meaningful decomposition quality with far fewer parameters (RQ1): the best SNN reached 0.811 macro correlation vs. TasNet’s 0.887, using $32\times$ fewer parameters. Their main limitation (RQ2) is reduced temporal capacity: they saturate earlier with data and are less reliable on weak or overlapping components, while DNNs pay in model size, training time, and inference latency. The choice depends on whether accuracy or deployability is the priority. The most effective modifications (RQ3) were residual TCN-style temporal modelling, attention, multi-scale branches, and per-component heads [1], [3], [10], all of which widen the temporal receptive field rather than merely adding parameters. Shallow models are not full replacements for DNNs, but are practical alternatives when compact, fast decomposition matters more than maximum accuracy.

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APPENDIX A DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

This declaration is made in accordance with the policy on the use of artificial intelligence in education set out by M.Dumontier [29]. During the preparation of this work, Grammarly was used to check and improve the grammar and spelling of the manuscript, and ChatGPT (OpenAI Codex) to assist with debugging and with documenting parts of the source code. These tools were not used to generate research results, design experiments, or produce scientific content. After using them, the thesis was carefully reviewed and edited the output and the author takes full responsibility for the content of this thesis.

APPENDIX B MODEL ARCHITECTURES

This appendix summarises the architecture of every model evaluated in the thesis. All models take a single-channel input of length 1024 and output K component channels of the same length ($K = 3$ for the baseline study, $K = 5$ for the extended study). Table IV lists the key hyperparameters and parameter counts; the prose below adds the structural detail.

Deep baselines. *TasNet* uses a 1-D convolutional encoder (128 filters, kernel 16, stride 8) followed by a temporal-convolutional separator of 3 repeats of 8 residual blocks each (bottleneck 128, hidden 256, skip 128, kernel 3, dilation doubling per block) with global layer normalisation, a sigmoid mask, and a learned 1-D transposed-convolution decoder. *UNet1D* uses a symmetric 1-D U-Net with base width 32: four downsampling stages (max-pool by 2, each a double 3-kernel convolution with batch-norm and ReLU) widening to a 512-channel bottleneck, four transposed-convolution upsampling stages with skip connections, and a 1×1 output convolution. *SepFormer* uses the same conv encoder/decoder as *TasNet* with a dual-path transformer separator: the latent sequence is chunked (size 100, hop 50) and processed by 2 *SepFormer* blocks, each containing an intra-chunk and an inter-chunk transformer encoder layer (model dim 128, 8 heads, feed-forward 256, GELU). *Shallow baselines and variants.* The *SingleHead MLP* flattens the input and maps it through a single fully connected layer to the output components, with no temporal structure. The *Classic Conv1D* base applies one Conv1d ($1 \rightarrow 32$, kernel 5, same padding) with ReLU and a 1×1 output head. The shallow variants extend this base by switching on the modules of Section II-B: *Multiscale Branches* adds three parallel branches with kernels $\{5, 7, 9\}$; *Multiscale Dilated* additionally sets branch dilations $\{1, 2, 4\}$; *Multiple-Head Multiscale Branches* gives each component its own head; *Fuse Shallow* adds 1×1 or bilinear fusion of the branches; the attention-stem variants prepend a kernel-9 convolutional stem with a lightweight attention weighting over branches. The *Autoencoder* uses a convolutional encoder with stride-2 downsampling at channel widths (16, 32, 64), kernel 5, and a symmetric transposed-convolution decoder with a 1×1 head. The strongest shallow model, *Attention-Stem Multi-Head*

TABLE IV
ARCHITECTURE AND PARAMETER COUNTS (3-COMPONENT
CONFIGURATION). MS = MULTI-SCALE BRANCHES, AS = ATTENTION
STEM, MH = MULTIPLE HEADS, RES = RESIDUAL CONNECTIONS.

Model	Core structure	Params
TasNet	conv enc/dec + TCN separator (3×8)	2,503,296
UNet1D	1-D U-Net, base 32, 4 levels	2,710,819
SepFormer	conv enc/dec + dual-path transformer	601,088
SingleHead MLP	flatten + 1 linear layer	643
Classic Conv1D	1 conv (k5) + 1×1 head	5,443
Multiscale Branches	3 branches (k 5/7/9)	5,859
Multiscale Dilated	branches + dilations 1/2/4	12,867
Multiple-Head Multiscale	MS + per-component heads	5,859
Autoencoder	conv AE, widths (16,32,64)	23,651
Fuse Shallow	MS + 1×1 /bilinear fusion	5,859
AS + MH + Multiscale Branches	AS + MS + MH	17,638
AS + Bilinear-Fusion + MH + MS	+ bilinear fusion	29,302
AS + MH + Multiscale TCN	AS + MS + MH + Res + TCN	78,070

Multiscale TCN (78,070 parameters), chains a kernel-9 stem, three multiscale branches (kernels $\{5, 7, 9\}$, dilations $\{1, 2, 4\}$, 16 channels each), 1×1 fusion to 48 channels, a 4-block TCN (48 channels, dilations $\{1, 2, 4, 8\}$) with residual connections, and K per-component output heads, using GroupNorm, GELU activations, and dropout 0.1.

APPENDIX C SHALLOW MODEL ARCHITECTURE EVOLUTION

Fig. 9 illustrates the six incremental stages by which the Classic Conv1D baseline was extended into the final *Attention-Stem Multi-Head Multiscale TCN*. Each row adds one or more components to the previous stage; components added at each step are highlighted with a dashed border.

APPENDIX D FULL LIST OF EVALUATION METRICS

Sixteen signal-level metrics were computed for every matched (prediction, ground-truth) component pair and then averaged. Table V gives each metric with its formula and a one-line interpretation. Here y and $\hat{y}$ are the ground-truth and predicted component, T the signal length, $Y = |\mathcal{F}(y)|$ and $\hat{Y} = |\mathcal{F}(\hat{y})|$ their magnitude spectra, and ϵ a small constant for numerical stability.

APPENDIX E HARD-CASE COMPONENT DECOMPOSITIONS

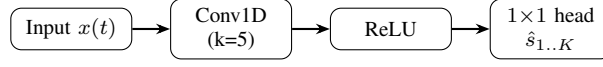
Figs. 10 and 11 show the full per-component decompositions for the representative hard case (the same 3-component mixture used in Section IV-E, containing two AM-FM components and one chirp). Each figure shows the input mixture, the summed reconstruction, and the three individually predicted components alongside their ground-truth counterparts. The shallow TCN (Fig. 10) recovers *amfm_1* with moderate fidelity but collapses almost completely on the low-amplitude *amfm_2*. *TasNet* (Fig. 11) also struggles with *amfm_2* but produces a larger predicted amplitude; neither model achieves a high correlation on this component (Table III).

TABLE V

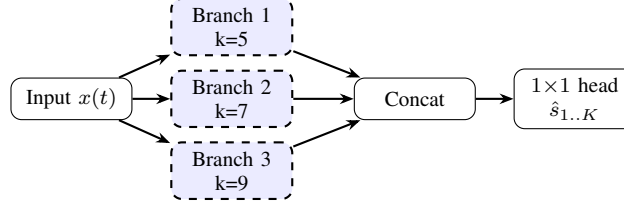
FULL LIST OF THE 16 EVALUATION METRICS. THE REMAINING METRICS ARE CONSISTENT WITH THESE AND ARE PROVIDED FOR COMPLETENESS.

Metric	Formula	Interpretation
MSE (test loss)	$\frac{1}{T} \sum_t (y_t - \hat{y}_t)^2$	Mean squared error; The direct training objective.
RMSE	$\sqrt{\text{MSE}}$	MSE on the original amplitude scale;
MAE	$\frac{1}{T} \sum_t y_t - \hat{y}_t $	Mean absolute error; less sensitive to outliers than MSE.
Normalised MSE	$\text{MSE} / (\frac{1}{T} \sum_t y_t^2 + \epsilon)$	MSE relative to signal energy; comparable across amplitudes.
Relative L_2 error	$\ y - \hat{y}\ _2 / (\ y\ _2 + \epsilon)$	Fractional energy of the error;
SNR (dB) (macro SNR)	$10 \log_{10} \frac{\sum_t y_t^2}{\sum_t (y_t - \hat{y}_t)^2}$	Signal-to-noise ratio; higher dB means smaller relative error.
PSNR (dB)	$20 \log_{10} \frac{\text{range}}{\sqrt{\text{MSE} + \epsilon}}$	Peak SNR relative to signal range; higher is better.
Correlation (macro corr.)	$\frac{\text{Cov}(y, \hat{y})}{\sigma_y \sigma_{\hat{y}}}$	Pearson correlation; shape similarity, scale-invariant, in $[-1, 1]$.
Cosine similarity	$\frac{\langle y, \hat{y} \rangle}{\ y\ \ \hat{y} \ }$	Angle between signals; 1 = identical direction.
SI-SDR (dB)	$10 \log_{10} \frac{\ \alpha y\ ^2}{\ \alpha y - \hat{y}\ ^2}, \alpha = \frac{\langle \hat{y}, y \rangle}{\ y\ ^2}$	Scale-invariant SDR; separation quality independent of gain.
Spectral convergence	$\ Y - \hat{Y}\ _2 / (\ Y\ _2 + \epsilon)$	Relative spectral error; lower means better frequency match.
Log-spectral distance	$\sqrt{\frac{1}{F} \sum_f (\log Y_f - \log \hat{Y}_f)^2}$	Distance between log-magnitude spectra; lower is better.
FFT magnitude L_1	$\frac{1}{F} \sum_f Y_f - \hat{Y}_f $	Mean absolute spectral error; the spectral training term.
Explained variance	$1 - \frac{\text{Var}(y - \hat{y})}{\text{Var}(y) + \epsilon}$	Fraction of variance captured; 1 is perfect, can be negative.
Energy ratio	$\frac{\sum_t \hat{y}_t^2 + \epsilon}{\sum_t y_t^2 + \epsilon}$	Predicted/true energy; > 1 over-predicts, < 1 under-predicts.
Max absolute error	$\max_t y_t - \hat{y}_t $	Worst-case pointwise deviation;

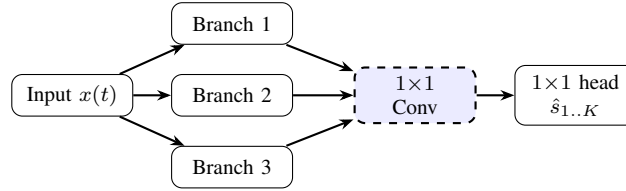
Stage 1: Classic Conv1D



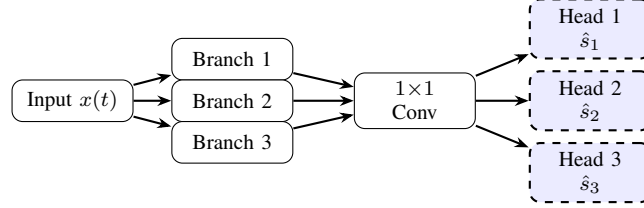
Stage 2: + Multi-scale branches



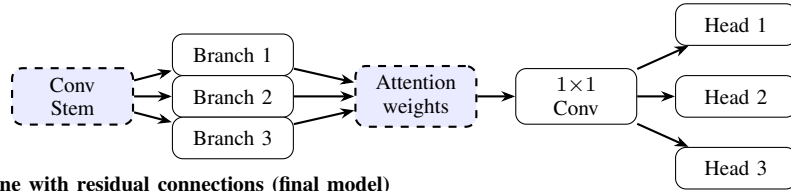
Stage 3: + Feature fusion (1x1 Conv)



Stage 4: + Multiple output heads



Stage 5: + Convolutional stem & attention weighting



Stage 6: + TCN backbone with residual connections (final model)

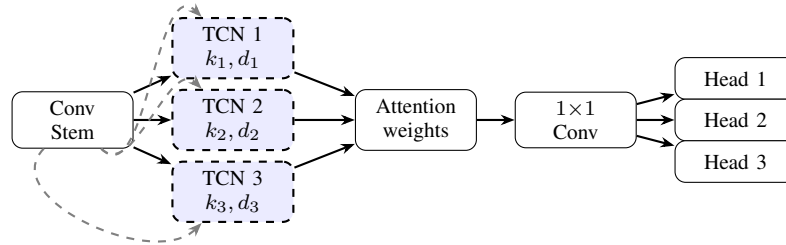


Fig. 9. Incremental evolution of the shallow SNN from the Classic Conv1D baseline (Stage 1) to the final *Attention-Stem Multi-Head Multiscale TCN* (Stage 6). Dashed-border blocks indicate components added at each stage. Stage 1: single Conv1D with one shared output head. Stage 2: single Conv1D replaced by three parallel branches with different kernel sizes. Stage 3: explicit 1x1 convolution fuses branch feature maps. Stage 4: single shared head replaced by per-component output heads. Stage 5: convolutional stem added before the branches; attention weighting selects informative scales. Stage 6: plain Conv1D branches upgraded to TCN blocks with dilated convolutions and residual connections (dashed grey arrows), forming the final model.

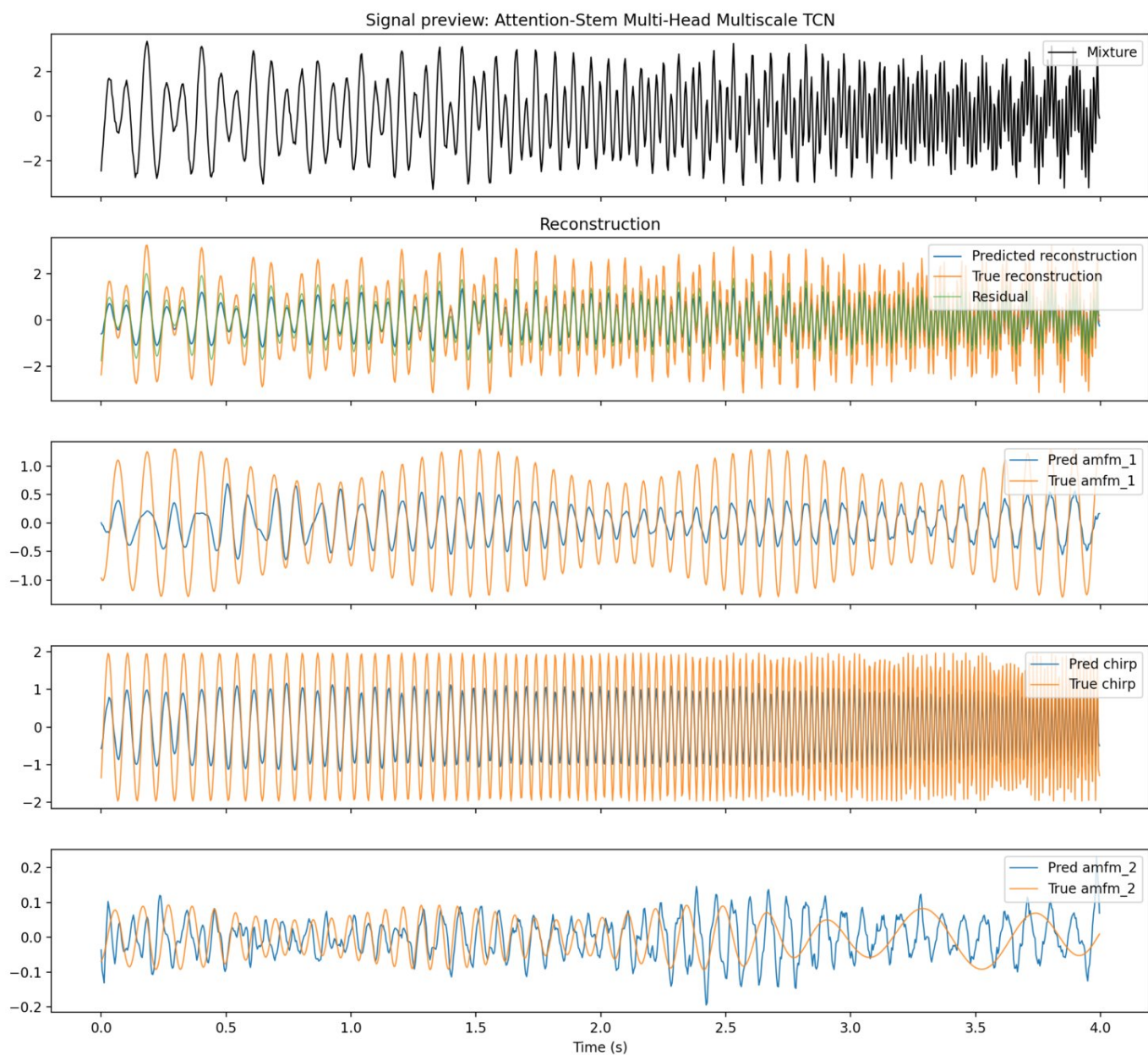


Fig. 10. Per-component decomposition of the hard-case mixture by the *Attention-Stem Multi-Head Multiscale TCN*. From top you can observe: input mixture, summed reconstruction with residual, and the three individual component predictions (blue) vs. ground truth (orange). The shallow model tracks amfm_1 and the chirp reasonably.

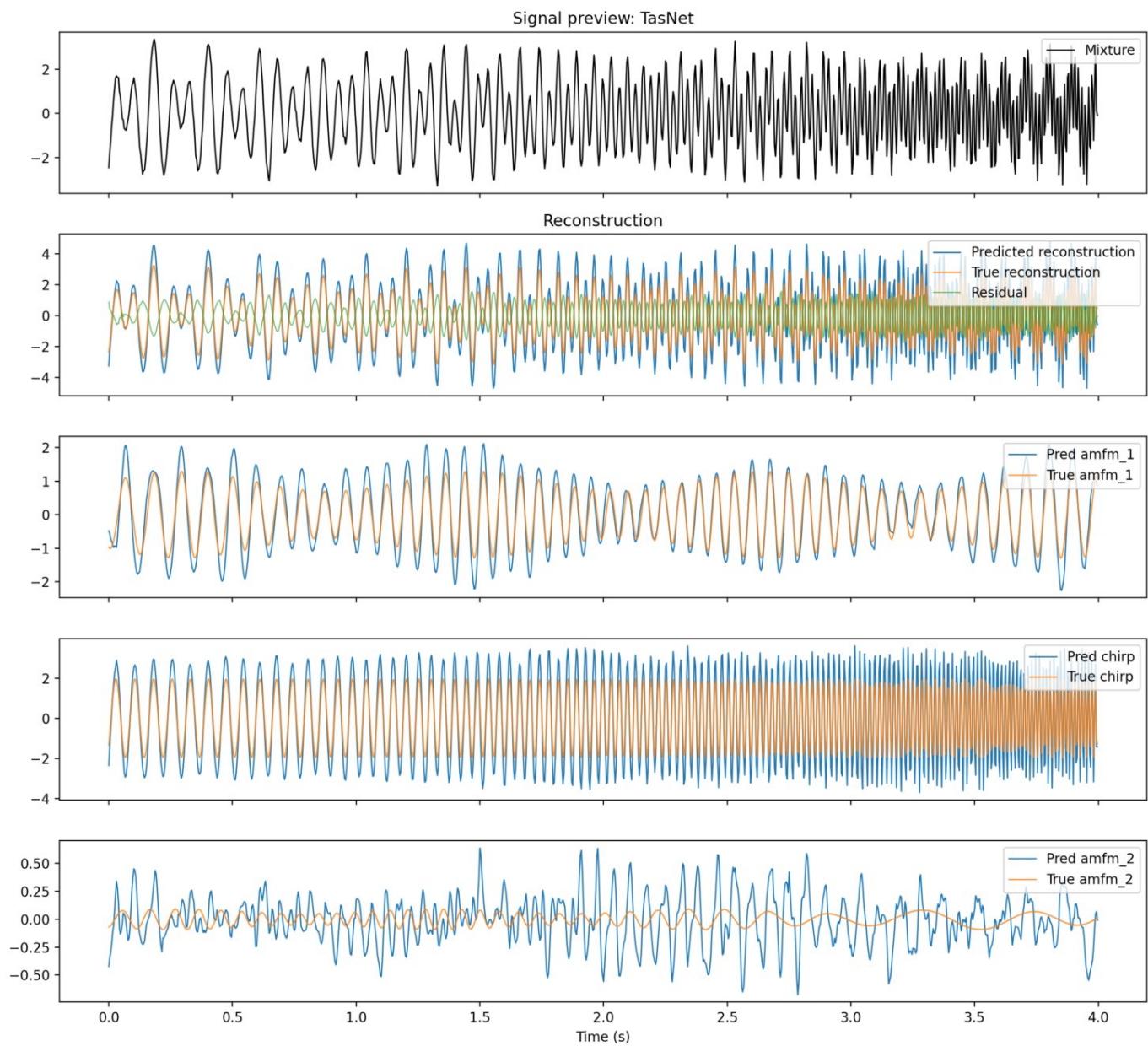


Fig. 11. Per-component decomposition of the same hard-case mixture by *TasNet*. *TasNet* follows the dominant *amfm_1* component closely but substantially overpredicts the amplitude of the chirp and *amfm_2*, consistent with the higher overprediction rate observed in the failure-label analysis (Fig. 7).